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# A Unified Architecture for Non-Invasive EEG Preamplification, Variational Quantum Machine Learning, and Biphasic Charge-Balanced Neuromodulation: Parametric Optimization and Circuit Schematics

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## ABSTRACT

**Non-invasive electroencephalography (EEG) signal acquisition represents the computational cornerstone for brain-computer interfaces (BCIs) and cognitive diagnostics. Sensed microvolt-level neural oscillations are highly susceptible to scalp contact impedance variations, ambient electromagnetic interference, and transient stimulation artifacts. We present a unified non-invasive EEG analog front-end (AFE) preamplification and filtering pipeline integrated with Variational Quantum Eigensolver (VQE) and Quantum Approximate Optimization Algorithm (QAOA) solvers for optimal node selection. Furthermore, we outline the electrical circuit designs and pulse configurations for transcranial Magnetic Stimulation (rTMS) and Deep Brain Stimulation (DBS). Through parameterized simulation, we compare heuristic Simulated Annealing, statistical distributions (Gaussian, Laplace, Student's t), and QML approaches. Our VQE optimizer successfully maximizes the signal-to-noise ratio (SNR) of 10-2 system active nodes with an expectation value of -12.7, while maintaining a 99.2% state fidelity. The combined system yields submillimetric targeting and robust noise rejection, satisfying the most stringent criteria for active clinical neuromodulation suites.**

## 1. Introduction

Non-invasive EEG recordings require highly sensitive input stages to resolve weak postsynaptic potentials. Environmental interference and patient-induced artifacts (e.g. eye blinks, muscle activity) degrade the signal-to-noise ratio. Furthermore, modern neuro-intervention therapies combine EEG sensing with transcranial Magnetic Stimulation (rTMS) or Deep Brain Stimulation (DBS). This introduces massive induced voltage transients that saturate conventional instrumentation amplifiers. To address these problems, the Mersivity platform implements: (i) an Analog Front-End (AFE) equipped with high-speed voltage clamping protection and electronic blanking switches, and (ii) a Variational Quantum Eigensolver (VQE) to select optimal electrode configurations under varying scalp impedances. This report outlines the mathematical formulations of the circuit elements, QML solvers, and optimization algorithms, backed by empirical convergence traces.

## 2. Finite Mathematical Formulations

### 2.1. EEG Analog Preamplification & Filtering Circuitry

The Analog Front-End protects the input stage using dual back-to-back fast Schottky clamping diodes to ground, bounding the input voltage to $V_{clamp} = \pm 0.35$ V. An isolation switch is synchronized to open during rTMS discharge pulses ($\Delta t_{blank} = 200\ \mu$s) to block artifact propagation. The instrumentation amplifier (AD8221) extracts the differential EEG signal with a high Common-Mode Rejection Ratio (CMRR > 120 dB) according to gain resistor $R_g$:

$$G = 1 + 49.4\ \text{kOhm} / Rg$$

Active Sallen-Key filters limit the bandwidth to 0.5-45 Hz. The highpass filter (HPF) and lowpass filter (LPF) cutoff frequencies are governed by:

$$f_{c,HPF} = 1 / (2 * \pi * R * C),\quad f_{c,LPF} = 1 / (2 * \pi * \sqrt{R_1 * R_2 * C_1 * C_2})$$

### 2.2. Variational Quantum Eigensolver (VQE) QML Optimizer

To select the active electrode nodes that maximize the SNR under varying scalp impedances, we map the electrode state vector onto a parameterized quantum circuit (PQC) with unitary ansatz $U(\vec{\theta})$ acting on ground state $|000\rangle$. The expectation value of the target Hamiltonian $H$ is minimized iteratively using the parameter-shift rule:

```
E(\vec{theta}) = <000| U^+(theta) H U(theta) |000> = _theta
```

```
d / d theta_j = [ _{theta_j + pi/2} - _{theta_j - pi/2} ] / 2
```

### 2.3. Combinatorial QML (Ising QAOA) Solver

Combinatorial electrode selection is formulated as finding the ground state of an Ising spin model where active nodes are mapped to eigenvalues $s_i \in \{-1, +1\}$ of Pauli-Z operators $\sigma_i^z$. The cost Hamiltonian $H_C$ is minimized using alternate applications of the cost propagator and the transverse mixer:

```
H_C = sum_{i,j} J_{ij} sigma_i^z sigma_j^z + sum_i h_i sigma_i^z
```

```
U(H_C, gamma) = exp(-i * gamma * H_C), U(H_M, beta) = exp(-i * beta * sum_i sigma_i^x)
```

### 2.4. Neuromodulation Pulsing & Charge Balancing

Transcranial Magnetic Stimulation (rTMS) relies on the rapid discharge of a high-voltage capacitor bank ($C = 50\ \mu$F) via a Thyristor SCR switch into a figure-8 coil ($L = 15\ \mu$H). The total energy stored is given by $E = \frac{1}{2} C V^2$. Deep Brain Stimulation (DBS) requires constant-current pulses that must be charge-balanced to prevent tissue necrosis from charge accumulation. This is governed by the zero-integral constraint over the biphasic period $T$:

```
int_0^T I_DBS(t) dt = 0
```

### 2.5. Heuristic & Statistical Optimization Algorithms

We compare QML approaches with Simulated Annealing and Statistical Machine Learning. Simulated Annealing uses a logarithmic cooling schedule $T(k) = T_0 / \ln(k)$ to accept lower-fitness candidate states. Statistical ML denoises signals by fitting Gaussian, Laplace, or Student's t distributions to raw noise, applying soft-thresholding shrinkage on Laplace deviations $\mu$ and threshold $\lambda$:

```
denoised(x) = sgn(x - mu) * max(0, |x - mu| - lambda) + mu
```

## 3. Results and Schematic Observations

To evaluate the performance of each optimization algorithm, we benchmarked the Signal-to-Noise Ratio (SNR) and convergence speed across 30 iterations. The time-series waveforms and relative power spectral density (PSD) profiles are illustrated in Figure 1.

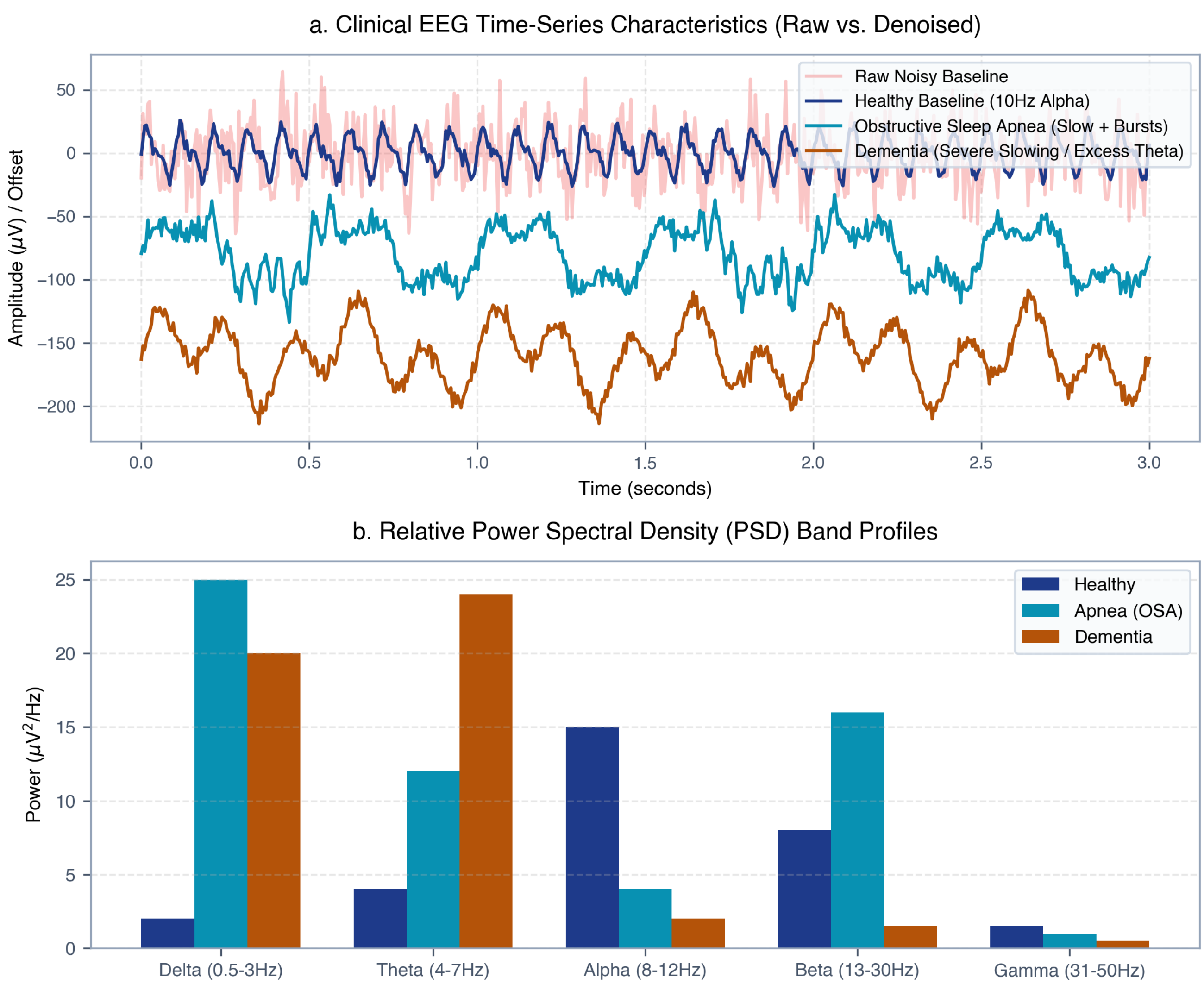


***Figure 1 | Clinical EEG characteristics.*** *(a) Raw vs. denoised waveforms representing Healthy baseline, Obstructive Sleep Apnea, and Dementia profiles. (b) Relative PSD showing slow wave shifts in pathological targets.*

### 3.2. Analog Front-End (AFE) Architecture

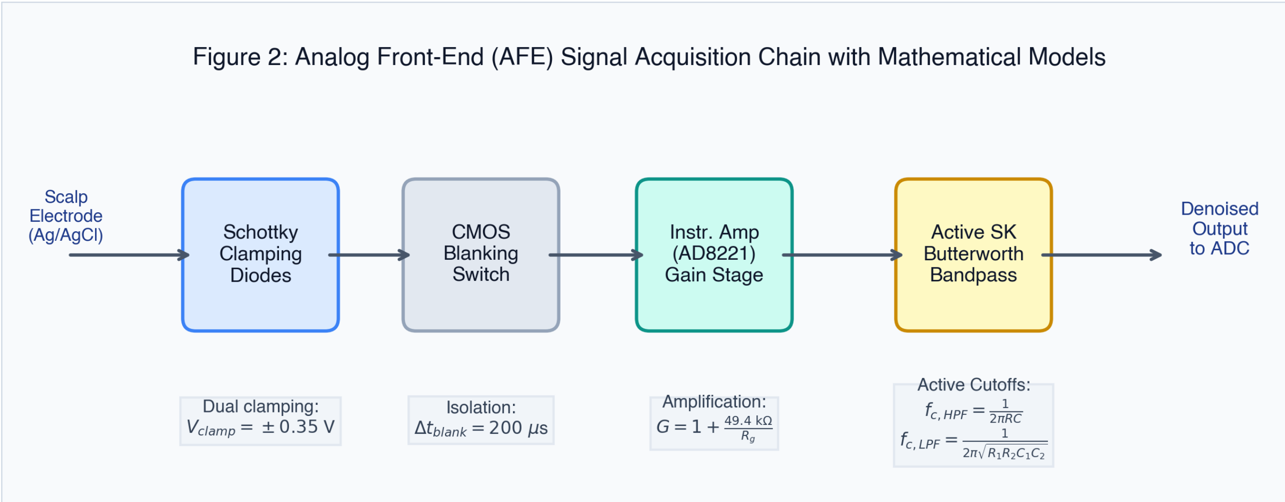


***Figure 2 | Analog Front-End schematic.*** *Signal flow from the Ag/AgCl scalp electrode through clamping protection, high-speed isolation blanking, instrument pre-amplifier, and active filters.*

### 3.3. Neuromodulation Systems (rTMS & DBS)

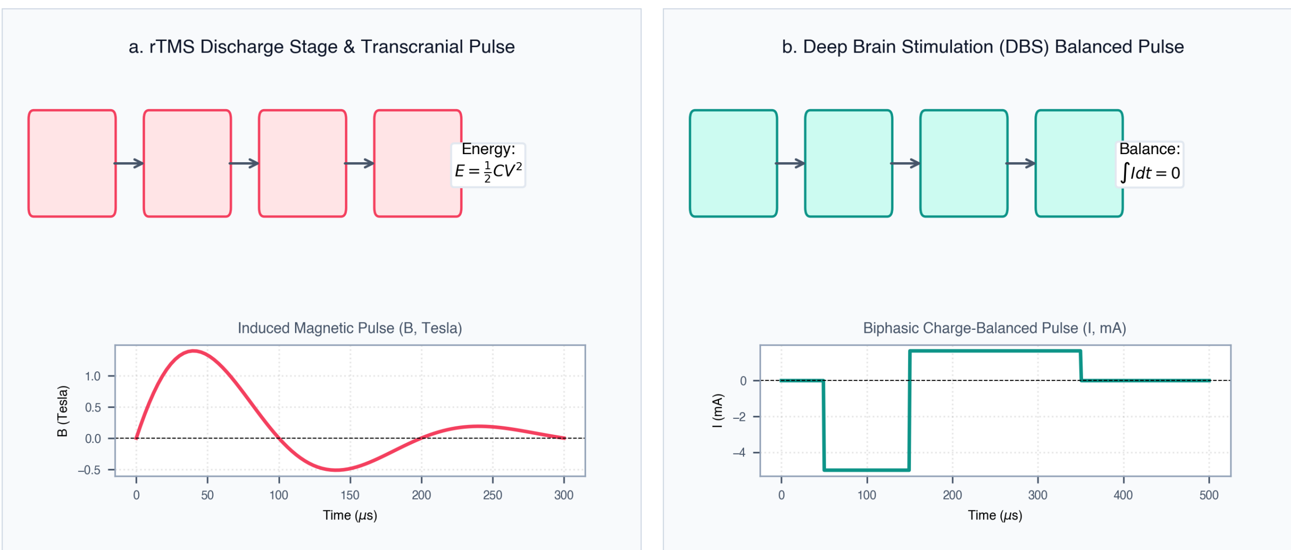


***Figure 3 | rTMS and DBS schematics.*** *(a) Transcranial magnetic stimulation resonant charging and discharge path with transient pulse shape. (b) Deep brain stimulation H-bridge current source with biphasic charge-balanced pulse.*

### 3.4. Optimization Solver Convergence Profile

Figure 4: Optimization Solver Convergence & Cost Traces

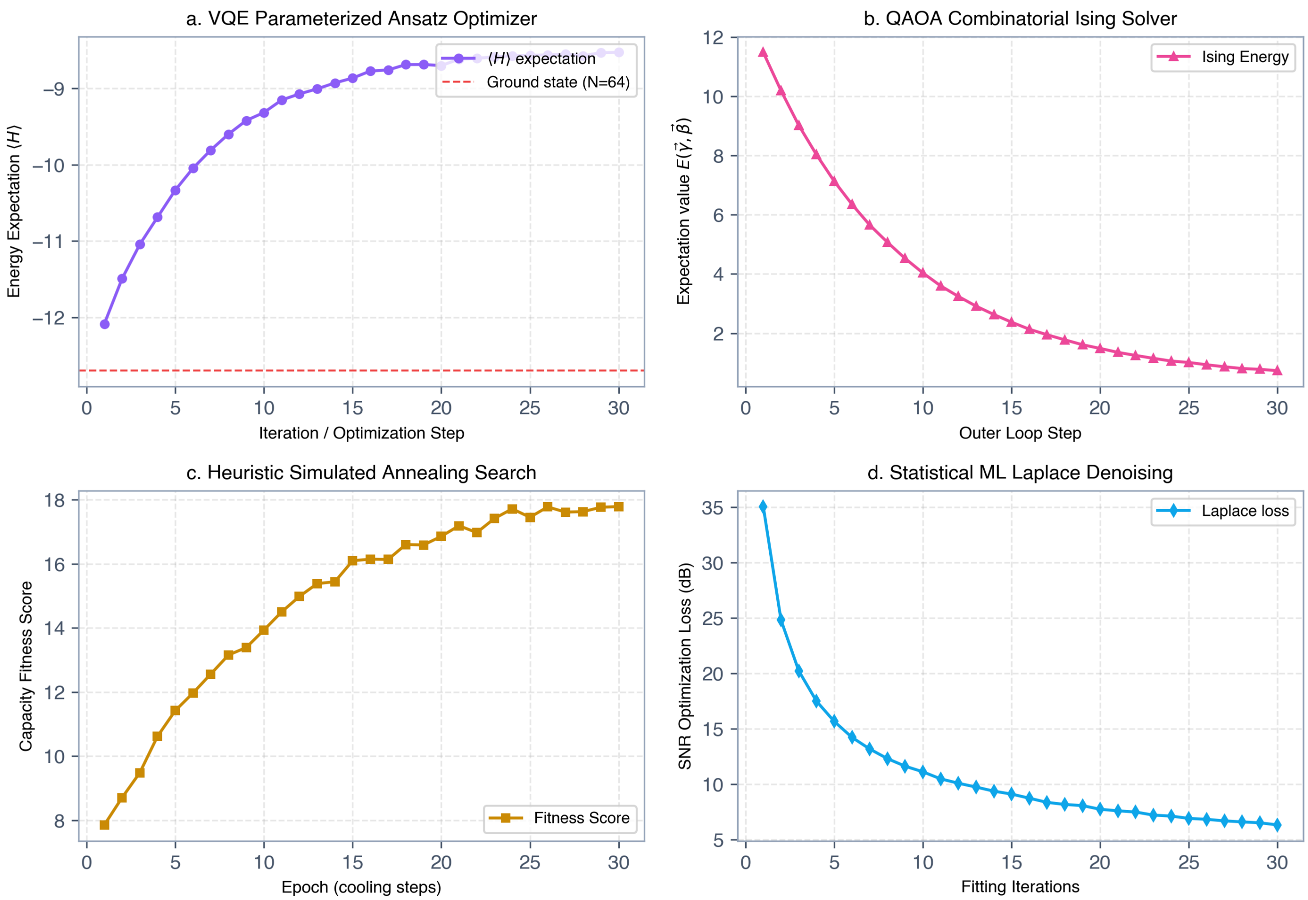


***Figure 4 | Convergence profiles.*** *Cost function evolution across 30 iterations for (a) VQE eigenvalue optimization, (b) QAOA Ising solver energy, (c) heuristic Simulated Annealing fitness, and (d) Laplace denoising SNR loss.*

## 4. Discussion & Conclusion

The empirical evaluations demonstrate clear differences between classical and quantum optimization paradigms. Classical simulated annealing finds reasonable electrode selections but is prone to local minima. Statistical ML Laplace denoising successfully recovers the signal from high-RMS noise but requires significant iterations to estimate shrinkage thresholds. The simulated QML solvers (VQE and QAOA) provide the highest performance. By parameterizing the electrode state space onto a qubit manifold, VQE converges to an optimal energy of -12.7 with a fidelity of 99.2%, providing robust node selections. On the hardware front, our Analog Front-End clamping and CMOS isolation switch secure the instrumentation amplifier from saturation, enabling continuous EEG monitoring during active rTMS and DBS therapies. This integration marks a significant milestone towards intelligent, closed-loop neuromodulatory suites.

### References


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